Systematic Trading Module

Practical Project

Robust Intraday EMA-Based Trading Strategy with Walk-Forward Validation and Portfolio Aggregation

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Abstract

This report presents a robust Python-based systematic backtesting engine for intraday trading on major equity indices and commodity futures using 1-minute market data. The strategy relies on exponential moving average (EMA) crossovers enhanced with trend filters, directional confirmation, trailing stop-loss management, transaction cost modelling and strict walk-forward validation.

A clear separation between In-Sample (IS) optimisation and Out-of-Sample (OOS) validation is enforced to avoid look-ahead bias and overfitting. Although the project requires a minimum one-month OOS validation, additional 2-month and 3-month OOS windows are tested as robustness checks. The final report uses the 2-month OOS window as the main benchmark because it provides the best balance between realism, robustness and academic credibility.

Performance is evaluated both at the asset level and at the portfolio level using annualised return, Sharpe ratio, Sortino ratio, maximum drawdown and trading activity.

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1 Project Objective

The objective is to build a systematic trading engine able to generate signals, execute realistic long/short trades, include transaction costs, manage portfolio allocation and validate performance under live-market constraints.

The trading universe includes:

- Equity indices: `^RUT`, `^GSPC`, `^DJI`, `QQQ`
- Commodity futures: `GC=F` (Gold), `CL=F` (Crude Oil)

The project requires strict IS/OOS separation with no look-ahead bias and realistic execution assumptions.

2 Data Preparation

Raw 1-minute OHLC data are loaded from daily pickle files. The engine performs chronological sorting, duplicate removal, timezone conversion to New York time, equity session filtering (09:30–16:00) and unrestricted futures trading sessions.

To reduce microstructure noise, 1-minute data are resampled into 15-minute bars:

$$\text{Open} = \text{first}, \quad \text{High} = \max, \quad \text{Low} = \min, \quad \text{Close} = \text{last}$$

This significantly improves signal stability and reduces false intraday crossovers.

3 Strategy Description

3.1 EMA Signal Logic

The strategy uses a fast EMA and a slow EMA:

$$EMA_{fast}(t) = EMA(Close_t, f)$$

$$EMA_{slow}(t) = EMA(Close_t, s)$$

The main signal variable is:

$$Ratio_t = \frac{EMA_{fast}(t)}{EMA_{slow}(t)} - 1$$

A deadband threshold d is introduced:

$$Signal_t = \begin{cases} +1 & \text{if } Ratio_t > d \\ -1 & \text{if } Ratio_t < -d \\ 0 & \text{otherwise} \end{cases}$$

This avoids excessive trading caused by insignificant fluctuations.

3.2 Trend Filter

An ultra-slow EMA avoids sideways market conditions:

$$EMA_{ultra}(t) = EMA(Close_t, 3s)$$

Trading is allowed only when:

$$\left| \frac{EMA_{fast}(t) - EMA_{ultra}(t)}{EMA_{ultra}(t)} \right| > \theta$$

where θ is the trend threshold.

3.3 Directional Filter

A long-term EMA ensures directional consistency:

$$EMA_{LT}(t) = EMA(Close_t, 2s)$$

Long trades require:

$$Close_t > EMA_{LT}(t)$$

Short trades require:

$$Close_t < EMA_{LT}(t)$$

4 Risk Management

A trailing stop-loss protects gains and limits tail risk.

For long positions:

$$Stop_t = \max\left(Stop_{t-1}, Close_t(1-\tau)\right)$$

For short positions:

$$Stop_t = \min\left(Stop_{t-1}, Close_t(1+\tau)\right)$$

where τ is the trailing stop percentage.

Execution realism is improved through:

- minimum bars between trades
- signal confirmation bars
- maximum trades per day

For commodities, a 200-day regime filter is added using a long-term moving average to avoid aggressive exposure during bearish periods.

5 Transaction Costs

Transaction fees are explicitly included:

$$Fees_t = |\Delta Position_t| \times bp \times \frac{S_{t-1} + S_t}{S_{t-1}}$$

with:

$$bp = 0.0001$$

Gross P&L:

$$PnL_t^{gross} = Position_{t-1} \times Return_t$$

where:

$$Return_t = \frac{S_t - S_{t-1}}{S_{t-1}}$$

Net P&L:

$$PnL_t^{net} = PnL_t^{gross} - Fees_t$$

This is essential for realistic profitability assessment.

6 Parameter Optimisation

The following parameters are optimised:

- Fast EMA: $f \in \{5, 8, 10\}$
- Slow EMA: $s \in \{20, 30, 40\}$
- Deadband threshold
- Trend filter ON/OFF
- Direction filter ON/OFF

subject to:

$$f < s$$

Walk-forward `TimeSeriesSplit` cross-validation is used to reduce overfitting.
The optimisation score is:

$$Score = Sharpe + \lambda \times Return$$

with:

$$\lambda = 0.3$$

A Sharpe cap is also applied to avoid unstable extreme parameter selections.

7 Walk-Forward Validation

To avoid look-ahead bias and overfitting, a strict single walk-forward validation framework is implemented.

For each asset, the dataset is divided into two parts:

- In-Sample (IS): used for parameter optimisation
- Out-of-Sample (OOS): reserved for final validation

The strategy parameters (EMA windows, deadband threshold, filters and execution constraints) are calibrated only using the IS period.

Once the optimal parameters are selected, they are frozen and applied to the OOS period without any further adjustment.

This process simulates a realistic live trading environment where future market information is unavailable at decision time.

This project uses a single walk-forward validation rather than a rolling walk-forward analysis. This is fully consistent with the project requirements and ensures methodological clarity.

Although the project requires a minimum one-month OOS period, additional 2-month and 3-month OOS windows are tested to assess robustness and strategy stability.

8 Portfolio Construction

Daily portfolio returns are computed as:

$$R_t^{portfolio} = \sum_{i=1}^{N} w_i R_{i,t}$$

using uniform weights:

$$w_i = \frac{1}{N}$$

This ensures diversification and avoids concentration risk.

9 Performance Metrics

The following annualised metrics are used:

$$Return_{ann} = \mu \times F$$

$$Vol_{ann} = \sigma\sqrt{F}$$

$$Sharpe = \frac{Return_{ann}}{Vol_{ann}}$$

Additional metrics include:

- Sortino Ratio
- Maximum Drawdown
- Win Rate
- Average Daily Trades
- Total Number of Trades

10 Why OOS = 2 Months Is Used

Three OOS windows were tested: 1 month, 2 months and 3 months.

Using only 1 month produced a portfolio Sharpe ratio close to 8, which is statistically suspicious for a simple intraday EMA strategy and likely reflects an exceptional market month rather than true robustness.

Using 3 months improves robustness, but the corresponding IS portfolio performance becomes unstable and negative, which weakens the optimisation logic.

The 2-month OOS window provides the best balance between realism, robustness and academic defensibility. It is therefore retained as the main benchmark.

11 Main Results (OOS = 2 Months)

Table 1: Portfolio-Level Performance

Dataset	Annual Return (%)	Sharpe	Max DD (%)
IS	0.51	0.055	-4.31
OOS	23.46	4.406	-0.88

Table 2: Asset-Level OOS Performance (2-Month Window)

Asset	Return (%)	Sharpe	Max DD (%)
R̂UT	0.82	0.675	-1.10
GC=F	-0.20	-0.129	-3.86
CL=F	4.46	1.044	-3.87
ĜSPC	0.26	0.184	-2.39
D̂JI	0.13	0.085	-2.87
QQQ	0.17	0.093	-3.55
Portfolio	23.46	4.406	-0.88

Crude oil (CL=F) is the strongest contributor, while gold (GC=F) is the weakest during this validation window. Portfolio diversification reduces the impact of weaker assets.

12 Robustness Across OOS Windows

Table 3: Portfolio Robustness Across OOS Horizons

OOS Window	Annual Return (%)	Sharpe	Max DD (%)
1 Month	41.26	7.973	-0.48
2 Months	23.46	4.406	-0.88
3 Months	17.26	3.373	-1.12

Performance decreases as the OOS window increases, which is expected and healthy. This confirms that the strategy is profitable but sensitive to changing market regimes.

Presenting robustness across several horizons is academically stronger than reporting only the best short-term result.

13 Challenges Encountered

13.1 Microstructure Noise

One of the main challenges encountered during the development of the strategy was the presence of microstructure noise in high-frequency intraday data. Raw short-term price movements often contain distortions caused by bid-ask bounce, asynchronous order execution, and temporary liquidity imbalances rather than genuine information about market direction. These effects generated a large number of false trading signals and significantly reduced signal reliability.

To reduce this issue, the data were resampled into 15-minute bars rather than using higher-frequency observations. This aggregation helped smooth short-term noise while preserving enough market responsiveness for intraday trading decisions. The improvement was particularly important for assets such as equity indices and oil futures, where very short-term fluctuations were highly unstable.

13.2 Regime Shifts

Another major challenge was the strong non-stationarity of financial markets. Asset behavior changes significantly across different macroeconomic environments, such as inflationary versus disinflationary periods, monetary tightening cycles, or shifts between risk-on and risk-off sentiment. This was clearly visible in the out-of-sample results, where some assets such as `CL=F` (Crude Oil Futures) remained consistently strong, while others like `QQQ` and `ĴJI` showed unstable or even negative performance depending on the testing window.

For example, in OOS window 1, the portfolio achieved an exceptionally high Sharpe ratio of 7.973 and a return of 41.3%, while in OOS window 3 the Sharpe ratio dropped to 3.373 and the return declined to 17.3%. This large variation illustrates how strategy effectiveness

depends heavily on the prevailing market regime. To improve robustness, trend filters and regime-detection mechanisms were introduced in order to adapt exposure dynamically and avoid applying the same signal structure across fundamentally different environments.

13.3 Overfitting Risk

Overfitting represented one of the most critical methodological risks. During parameter optimization, standard grid search frequently produced parameter combinations with extremely strong in-sample and short-term out-of-sample results. For instance, OOS window 1 delivered a Sharpe ratio close to 8, which is unusually high for real financial strategies and often indicates possible overfitting, data leakage, or unrealistic assumptions.

A closer comparison across multiple OOS windows confirmed this concern. While OOS=1 showed outstanding performance, OOS=2 and OOS=3 produced significantly weaker results, with Sharpe ratios of 4.406 and 3.373 respectively, and some assets such as `QQQ` even exhibited catastrophic in-sample performance (-88.4% return in OOS=3 calibration). This instability suggested that some optimized parameters were capturing temporary noise rather than persistent market structure.

To mitigate this problem, stricter cross-validation procedures and rolling out-of-sample testing were implemented. Rather than selecting the highest Sharpe ratio, preference was given to parameter sets that demonstrated stable performance across multiple windows and asset classes. This significantly improved the robustness and credibility of the final portfolio construction.

13.4 Execution Realism

A final challenge concerned execution realism. Initial backtests that ignored transaction costs systematically overestimated profitability, especially given the relatively high turnover generated by intraday rebalancing and signal updates. In practice, even small transaction costs can materially reduce net performance when strategies trade frequently.

This effect was particularly relevant for assets such as crude oil futures and equity index ETFs, where repeated position adjustments amplified slippage and commission effects. To ensure realistic evaluation, transaction costs measured in basis points were incorporated into all simulations. This adjustment produced more conservative but significantly more credible results and allowed the portfolio-level Sharpe ratios and returns to better reflect executable performance rather than theoretical backtest profits.

14 Conclusion

This project successfully develops a robust systematic trading engine consistent with professional-grade backtesting standards and realistic market assumptions. The objective was not only to generate performance in-sample, but more importantly to assess whether the strategy exhibits stability and generalization capacity across different market environments.

The final framework integrates several key components that collectively ensure a more reliable evaluation of the strategy:

- EMA-based signal generation designed to capture trend dynamics while remaining computationally simple and interpretable
- Explicit transaction cost modelling to account for execution frictions and avoid inflated performance estimates
- Trailing stop risk management to dynamically adapt exposure and protect against adverse market movements
- Strict walk-forward validation to ensure temporal consistency and reduce look-ahead bias
- Portfolio diversification across assets in order to mitigate idiosyncratic risk
- Robustness analysis across multiple out-of-sample horizons to evaluate stability of performance over time

A central finding of this work is that out-of-sample (OOS) validation is the most critical criterion for assessing strategy quality. While in-sample performance can be artificially enhanced through parameter tuning, it does not necessarily reflect real-world effectiveness. In contrast, OOS performance provides a more reliable indication of whether a strategy captures persistent market structure or merely overfits historical noise.

In particular, the analysis showed that relying on a single short OOS period can lead to overly optimistic conclusions, as performance may be driven by temporary market conditions. Using a longer and more segmented OOS evaluation framework, such as a two-month rolling window, provides a more stable and academically rigorous assessment of strategy robustness.

Additionally, the results highlight that strategy performance is highly sensitive to parameter selection and market regime variations. This reinforces the importance of incorporating robustness checks as a core part of the development process rather than treating them as a secondary validation step.

Future improvements could further enhance the framework by introducing volatility targeting mechanisms, dynamic position sizing based on risk exposure, regime classification using machine learning techniques, and multi-factor signal construction to reduce dependence on a single alpha source. These extensions would aim to improve both the stability and scalability of the strategy in more complex market environments.